

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

```
data=pd.read_csv('/chicago_taxi_trips_2016_01.csv')
```

```
data.head()
```

	taxi_id	trip_start_timestamp	trip_end_timestamp	trip_seconds	trip_miles	pickup_census_tract	dropoff_census_tract	pickup_community_area	dropoff_community_area
0	85.0	2016-1-13 06:15:00	2016-1-13 06:15:00	180.0	0.40	NaN	NaN	NaN	NaN
1	2776.0	2016-1-22 09:30:00	2016-1-22 09:45:00	240.0	0.70	NaN	NaN	NaN	NaN
2	3168.0	2016-1-31 21:30:00	2016-1-31 21:30:00	0.0	0.00	NaN	NaN	NaN	NaN
3	4237.0	2016-1-23 17:30:00	2016-1-23 17:30:00	480.0	1.10	NaN	NaN	NaN	NaN
4	5710.0	2016-1-14 05:45:00	2016-1-14 06:00:00	480.0	2.71	NaN	NaN	NaN	NaN

```
data.shape
```

```
(77591, 20)
```

```
data.isnull().sum()
```

	0
taxi_id	2
trip_start_timestamp	0
trip_end_timestamp	8
trip_seconds	22
trip_miles	0
pickup_census_tract	77591
dropoff_census_tract	33623
pickup_community_area	13026
dropoff_community_area	14236
fare	2
tips	2
tolls	2
extras	3
trip_total	3
payment_type	1
company	28876
pickup_latitude	13026
pickup_longitude	13026
dropoff_latitude	14140
dropoff_longitude	14140

```
dtype: int64
```

```
data_cleaned = data[['trip_miles', 'trip_total']].dropna()
x = data_cleaned[['trip_miles']].values
y = data_cleaned[['trip_total']].values
print(x)
print(y)
```

```
[[0.4 ]
 [0.7 ]
 [0.  ]
 ...
 [1.05]
 [1.1 ]
 [0.8 ]]
[ 4.5   8.9  47.75 ...  8.75 10.5   5.5 ]
```

```
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2, random_state = 42)
```

```
model=LinearRegression()
model.fit(x_train, y_train)
y_pred=model.predict(x_test)
```

```
print(y_pred)
print(y_test)
```

```
[15.40526111 15.17887659 16.10499505 ... 16.002093  15.24061782
 15.15829618]
[10.25  5.5  17.7   ... 18.3   4.75  8.5 ]
```

```
plt.scatter(x_train, y_train, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (training set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```



```
plt.scatter(x_test, y_test, color='blue')
plt.plot(x_train, model.predict(x_train), color='red')
plt.title('SALARY VS EXPERIENCE (testing set)')
plt.xlabel('Experience in Years')
plt.ylabel('Salary in Rupees')
plt.show()
```

